

Image Transformation from Polar to Cartesian Coordinates Simplifies the Segmentation of Brain Images

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Abstract — A new image transformation has been designed to simplify the segmentation of the MR and CT brain images. This transformation supposes that the head is a cylindrical object and can be modeled more efficiently in polar coordinates, which gives a new set of geometrically constant image features. For example, the air-skin surface can be modeled with a one-dimensional function. An image transformation from the polar coordinate system to the Cartesian coordinate system has been used, because the algorithm realization and the actual image processing are easier and faster in Cartesian coordinates and the results equal to polar coordinate processing.

I. Introduction

Many methods have been presented to segment the brain. Image pixels can be classified by measurement values in multiecho images [1], but single-echo images require a rule-based approach [2]. However, no reliable algorithms have been found to provide fully automatic segmentation of single-echo images. Only few segmentation algorithms take account the shape of the head and the brain. It also seems to be difficult to use advanced filtering for high level segment boundaries to make the segmentation reliable.

Our goal in brain image segmentation is three-dimensional intra-operative visualization of brain anatomy. The visualization needs three different segments: background, head and brain. The segmentation process should be fully automatic for clinical use. [3]

II. Methods

The segmentation method presented in this paper was developed from the idea of modeling the head as a cylindrical object. This should simplify mathematical modeling of segment boundaries and enable form based segmentation and high level boundary analysis.

The segmentation is done in phases. The image topology is first transformed from the polar to the Cartesian coordinates. Then the skin is filtered away. Erosion and dilation are applied to the data to remove bone. The result is transformed back to the original coordinate system and used as a preliminary segment for the original image. A Mexican-hat filtering is then used with the orig-

inal image and the preliminary segment to classify all the pixels.

A. Image transformation

This transformation is done independently for each image slice. The transformed image $T(x, y)$ is generated pixel by pixel from the original image $I(x_i, y_i)$ using (1) (Fig. 1). This transformation is dependent on a selectable mid-point (x_m, y_m) of the image. The size (x_s, y_s) of the transformed image is also selectable.

$$T(x, y) = I(y \cos \frac{2\pi x}{x_s} + x_m, y \sin \frac{2\pi x}{x_s} + y_m). \quad (1)$$

The transformed image should have a larger total area than the original. Otherwise some information is lost in the transformation and the transformation is irreversible.

B. Removing the skin

First an algorithm tries to find the air-skin boundary from a transformed image. The skin-bone boundary should be next and the bone-brain boundary just below the skin-bone boundary. When a boundary has been generated, it is analyzed. If any major discontinuities are found, the boundary will be interpolated over these.

The methods for finding different boundaries in the images vary with the origin of the images. It is easy to detect bone from CT scans, but much more difficult from MRI. Some bone could be classified as brain by the bone-brain boundary detection algorithm. Other methods are needed to filter the bone.

C. Gray level erosion and dilation

Erosion and dilation can be used to separate objects connected with narrow links [4]. Some iterations of erosion filter bone in the MR images, because bone is noisy and faint. The same amount of gray level dilation restores the shape of the brain nearly to the original. We have used erosion and dilation with 3x3 neighbourhood from one to three iterations depending on imaging parameters and pixel size.

D. Inverse image transformation

The original topology is restored using (1) for each image pixel (Fig. 2). The result is an approximate segmentation, a preliminary segment of the brain, and it can be used in further segmentation.

E. Mexican-hat filtering

The Mexican-hat filter has been used to gain information about the inner structure of the brain. The pixels in the preliminary segment are filtered with the Mexican-hat filter, which generates neighbourhood information. Thresholding the output of the Mexican-hat filtering to binary values can be avoided by using fuzzy logic in further analysis.

F. Analyzing

The Mexican-hat filtering produces negative values for the brain, because brain is "darker" than CSF in T2 weighted images. However, very dark pixels, which easily get negative Mexican-hat values, do not belong to the brain. A fuzzy rule was applied: "All the medium gray pixels with a negative Mexican-hat value belong to the brain." When the fuzzy rule is applied for all the pixels in the preliminary segment, the brain is segmented. A segmented slice and a three dimensional reconstruction can be seen in Fig. 3.

III. Results and Conclusions

Our results show that the use of polar coordinates simplifies the boundary classification. It is easier to model the air-skin, skin-bone and bone-brain boundaries. The usability of the image transformation from the polar to Cartesian coordinate system has also been tested.

We will test the effects of orientation dependent filtering in polar coordinates in the near future. Proper filtering for the transformed image should enhance the boundary classification.

The method presented here can segment the brain from a noisy image with spatial intensity variations. The algorithms do not rely much on the absolute gray level values, but more on the topology and gray level relations of the image.

References

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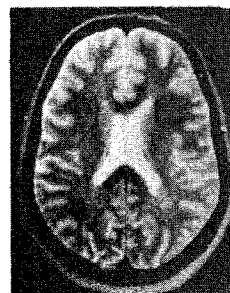


Fig. 1. Image transformation from polar to Cartesian coordinates

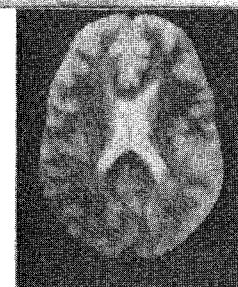


Fig. 2. Inverse image transformation from Cartesian to polar coordinates after the skin has been removed

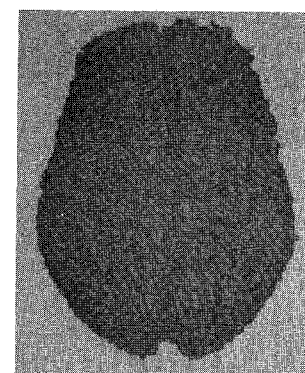


Fig. 3. Segmented MRI slice and a three dimensional reconstruction from segmented slices